

RNA polymerase II depletion from the inactive X chromosome territory is not mediated by physical compartmentalization.

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Subnuclear compartmentalization has been proposed to play an important role in gene regulation by segregating active and inactive parts of the genome in distinct physical and biochemical environments. During X chromosome inactivation (XCI), the noncoding Xist RNA coats the X chromosome, triggers gene silencing and forms a dense body of heterochromatin from which the transcription machinery appears to be excluded. Phase separation has been proposed to be involved in XCI, and might explain the exclusion of the transcription machinery by preventing its diffusion into the Xist-coated territory. Here, using quantitative fluorescence microscopy and single-particle tracking, we show that RNA polymerase II (RNAPII) freely accesses the Xist territory during the initiation of XCI. Instead, the apparent depletion of RNAPII is due to the loss of its chromatin stably bound fraction. These findings indicate that initial exclusion of RNAPII from the inactive X reflects the absence of actively transcribing RNAPII, rather than a consequence of putative physical compartmentalization of the inactive X heterochromatin domain.

A Pooled shRNA Screen Identifies Rbm15, Spen, and Wtap as Factors Required for Xist RNA-Mediated Silencing.

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X-chromosome inactivation is the process that evolved in mammals to equalize levels of X-linked gene expression in XX females relative to XY males. Silencing of a single X chromosome in female cells is mediated by the non-coding RNA Xist. Although progress has been made toward identifying factors that function in the maintenance of X inactivation, the primary silencing factors are largely undefined. We developed an shRNA screening strategy to produce a ranked list of candidate primary silencing factors. Validation experiments performed on several of the top hits identified the SPOC domain RNA binding proteins Rbm15 and Spen and Wtap, a component of the m6A RNA methyltransferase complex, as playing an important role in the establishment of Xist-mediated silencing. Localization analysis using super-resolution 3D-SIM microscopy demonstrates that these factors co-localize with Xist RNA within the nuclear matrix subcompartment, consistent with a direct interaction.

B cell-specific XIST complex enforces X-inactivation and restrains atypical B cells.

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The long non-coding RNA (lncRNA) XIST establishes X chromosome inactivation (XCI) in female cells in early development and thereafter is thought to be largely dispensable. Here, we show XIST is continually required in adult human B cells to silence a subset of X-linked immune genes such as TLR7. XIST-dependent genes lack promoter DNA methylation and require continual XIST-dependent histone deacetylation. XIST RNA-directed proteomics and CRISPRi screen reveal distinctive somatic cell-type-specific XIST complexes and identify TRIM28 that mediates Pol II pausing at promoters of X-linked genes in B cells. Single-cell transcriptome data of female patients with either systemic lupus erythematosus or COVID-19 infection revealed XIST dysregulation, reflected by escape of XIST-dependent genes, in CD11c+ atypical memory B cells (ABCs). XIST inactivation with TLR7 agonism suffices to

promote isotype-switched ABCs. These results indicate cell-type-specific diversification and function for lncRNA-protein complexes and suggest expanded roles for XIST in sex-differences in biology and medicine.

X-inactivation: quantitative predictions of protein interactions in the Xist network.

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The transcriptional silencing of one of the female X-chromosomes is a finely regulated process that requires accumulation in cis of the long non-coding RNA X-inactive-specific transcript (Xist) followed by a series of epigenetic modifications. Little is known about the molecular machinery regulating initiation and maintenance of chromosomal silencing. Here, we introduce a new version of our algorithm catRAPID to investigate Xist associations with a number of proteins involved in epigenetic regulation, nuclear scaffolding, transcription and splicing processes. Our method correctly identifies binding regions and affinities of protein interactions, providing a powerful theoretical framework for the study of X-chromosome inactivation and other events mediated by ribonucleoprotein associations.

The prognostic potential of long noncoding RNA XIST in cardiovascular diseases: a review.

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There is a significant mortality rate associated with cardiovascular disease despite advances in treatment. long Non-coding RNAs (lncRNAs) play a critical role in many biological processes and their dysregulation is associated with a wide range of diseases in which their downstream pathways are disrupted. A lncRNA X-inactive specific transcript (XIST) is well known as a factor that regulates the physiological process of chromosome dosage compensation for females. According to recent studies, lncRNA XIST is involved in a variety of cellular processes, including apoptosis, proliferation, invasion, metastasis, oxidative stress and inflammation, through molecular networks with microRNAs and their downstream targets in neoplastic and non-neoplastic diseases. Because these cellular processes play a role in the pathogenesis of cardiovascular diseases, we aim to investigate the role that lncRNA XIST plays in this process. Additionally, we wish to determine whether it is a prognostic factor or a potential therapeutic target in these diseases.

Long non-coding RNAs: spatial amplifiers that control nuclear structure and gene expression.

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Over the past decade, it has become clear that mammalian genomes encode thousands of long non-coding RNAs (lncRNAs), many of which are now implicated in diverse biological processes. Recent work studying the molecular mechanisms of several key examples - including Xist, which orchestrates X chromosome inactivation - has provided new insights into how lncRNAs can control cellular functions by acting in the nucleus. Here we discuss emerging mechanistic insights into how lncRNAs can regulate

gene expression by coordinating regulatory proteins, localizing to target loci and shaping three-dimensional (3D) nuclear organization. We explore these principles to highlight biological challenges in gene regulation, in which lncRNAs are well-suited to perform roles that cannot be carried out by DNA elements or protein regulators alone, such as acting as spatial amplifiers of regulatory signals in the nucleus.

LncRNA-XIST Promotes Proliferation and Migration in ox-LDL Stimulated Vascular Smooth Muscle Cells through miR-539-5p/SPP1 Axis.

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Long noncoding RNAs (lncRNAs) are untranslated transcripts greater than 200 nucleotides in length. Despite not being translated, they play a role in the regulation of transcription, translation, and other cellular processes and have been identified as key regulator in the progression of atherosclerosis. This study focused on the lncRNA X-inactive specific transcript (XIST), which participates in the regulation of X chromosome inactivation. XIST is produced by the XIST gene and is located on human chromosome Xq13.2. We also focused on discovering the possible role and mechanism of lncRNA XIST in oxidized low-density lipoprotein- (ox-LDL-) stimulated vascular smooth muscle cells (VSMCs), which could further help evaluate its possible role in the progression of atherosclerosis. XIST was overexpressed in ox-LDL-stimulated VSMCs, while the expression of miR-539-5p was decreased. XIST knockdown hindered the proliferation and migration of ox-LDL-treated VSMCs. XIST inhibits the miR-539-5p expression through direct interaction. Besides, miR-539-5p inhibitors can partially reverse the effect of XIST depletion on the proliferation and migration of VSMCs induced by ox-LDL stimulation. Further mechanistic analysis showed that secreted phosphoprotein 1 (SPP1) is the target of miR-539-5p, and XIST acts as a competing endogenous RNA for miR-539-5p to enhance the expression of SPP1. In addition, miR-539-5p inhibitor exerts its proliferation and migration effects by activating the miR-539-5p/SPP1 axis in VSMCs stimulated by ox-LDL. In conclusion, our study findings show that XIST inhibition can inhibit the proliferation and migration of atherosclerosis vascular smooth muscle cells, which provides a new theoretical basis for atherosclerosis treatment.

How does the Xist activator Rlim/Rnf12 regulate Xist expression?

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The long non-coding RNA (lncRNA) Xist is crucially involved in a process called X chromosome inactivation (XCI), the transcriptional silencing of one of the two X chromosomes in female mammals to achieve X dosage compensation between the sexes. Because Xist RNA silences the X chromosome from which it is transcribed, the activation of Xist transcription marks the initiation of the XCI process and thus, mechanisms and players that activate this gene are of central importance to the XCI process. During female mouse embryogenesis, XCI occurs in two steps. At the 2-4 cell stages imprinted XCI (iXCI) silences exclusively the paternally inherited X chromosome (Xp). While extraembryonic cells including trophoblasts keep the Xp silenced, epiblast cells that give rise to the embryo proper reactivate the Xp and undergo random XCI (rXCI) around implantation. Both iXCI and rXCI are dependent on Xist. Rlim, also known as Rnf12, is an X-linked E3 ubiquitin ligase that is involved in the transcriptional activation of Xist. However, while data on the crucial involvement of Rlim during iXCI appear clear, its role in rXCI has been

controversial. This review discusses data leading to this disagreement and recent evidence for a regulatory switch of Xist transcription in epiblasts of implanting embryos, partially reconciling the roles of Rlim during Xist activation.

In vivo ultraviolet and dimethyl sulfate footprinting of the 5' region of the expressed and silent Xist alleles.

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The Xist (X inactive specific transcript) gene plays an essential role in X chromosome inactivation. To elucidate the mechanisms controlling Xist expression and X inactivation, we examined in vivo DNA-protein interactions in the Xist promoter region in a female mouse cell line (BMSL2), which has distinguishable Xist alleles. In vivo footprinting was accomplished by treatment of cells with dimethyl sulfate or ultraviolet light, followed by ligation-mediated polymerase chain reaction of purified DNA. The expressed allele on the inactive X chromosome and the silent allele on the active X chromosome were separated by the use of a restriction fragment length polymorphism prior to ligation-mediated polymerase chain reaction. The chromatin structure of the Xist promoter was found to be consistent with the activity state of the Xist gene. The silent allele (on the active X chromosome) showed no footprints, while the expressed allele (on the inactive X chromosome) showed footprints at a consensus sequence for a CCAAT box, two weak Sp1 sites, and a weak TATA box.

X-inactive-specific transcript: a long noncoding RNA with a complex role in sex differences in human disease.

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In humans, the X and Y chromosomes determine the biological sex, XX specifying for females and XY for males. The long noncoding RNA X-inactive specific transcript (lncRNA XIST) plays a crucial role in the process of X chromosome inactivation (XCI) in cells of the female, a process that ensures the balanced expression of X-linked genes between sexes. Initially, it was believed that XIST can be expressed only from the inactive X chromosome (Xi) and is considered a typically female-specific transcript. However, accumulating evidence suggests that XIST can be detected in male cells as well, and it participates in the development of cancers and other human diseases by regulating gene expression at epigenetic, chromatin remodeling, transcriptional, and translational levels. XIST is abnormally expressed in many sexually dimorphic diseases, including autoimmune and neurological diseases, pulmonary arterial hypertension (PAH), and some types of cancers. However, the underlying mechanisms are not fully understood. Escape from XCI and skewed XCI also contributes to sex-biased diseases and their severity. Interestingly, in humans, similar to experimental animal models of human disease, the males with the XIST gene activated display the sex-biased disease condition at a rate close to females, and significantly greater than males who had not been genetically modified. For instance, the men with supernumerary X chromosomes, such as men with Klinefelter syndrome (47, XXY), are predisposed toward autoimmunity similar to females (46, XX), and have increased risk for strongly female biased diseases, compared to 46, XY males. Interestingly, chromosome X content has been linked to a longer life span, and the presence of two chromosome X contributes to increased longevity regardless of the hormonal status. In this review, we summarize recent knowledge about XIST structure/function correlation and involvement in human

disease with focus on XIST abnormal expression in males. Many human diseases show differences between males and females in penetrance, presentation, progression, and survival. In humans, the X and Y sex chromosomes determine the biological sex, XX specifying for females and XY for males. This numeric imbalance, two X chromosomes in females and only one in males, known as sex chromosome dosage inequality, is corrected in the first days of embryonic development by inactivating one of the X chromosomes in females. While this "dosage compensation" should in theory solve the difference in the number of genes between sexes, the expressed doses of X genes are incompletely compensated by X chromosome inactivation in females. In this review we try to highlight how abnormal expression and function of XIST, a gene on the X chromosome responsible for this inactivation process, may explain the sex differences in human health and disease. A better understanding of the molecular mechanisms of XIST participation in the male-female differences in disease is highly relevant since it would allow for improving the personalization of diagnosis and sex-specific treatment of patients.
